# The empirical recurrence for OEIS A263245, proved from an exact model

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**Abstract**

OEIS A263245 carries an empirical recurrence of order 3 contributed by Colin Barker. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a certified growth pattern of the automaton's row word, from which $a$ provably satisfies a MONIC linear recurrence of order $S = 5$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates $a$ is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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## 1 The conjecture

OEIS A263245 is "Decimal representation of the n-th iteration of the "Rule 155" elementary cellular automaton starting with a single ON (black) cell.". It has offset 0 and begins

$$1, 5, 19, 95, 319, 1535, 5119, 24575, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Conjectures from _Colin Barker_, Jan 20 2016 and Apr 17 2019: (Start) a(n) = a(n-1)+16*a(n-2)-16*a(n-3) for n>3.

As of the "Last modified" line on the live entry (Feb 16 2025 08:33:27, revision 31) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 The value is a certified growth

The entry reads a row of the automaton at stage $n$ as a string of digits and takes its numeric value. Nothing is being counted, so there is no digraph. What settles the sequence is that the string itself grows by a fixed pattern: the automaton was run and the certificate

$$w(n+p) \ = \ L_r + w(n) + R_r$$

was verified for every available $n$ in each residue class $r$ of $n$ modulo the period $p$, after a pre-period $n_0$ during which the shape has not yet settled. Concatenation is multiplication by a power of the base $B$ and addition, so along a class

$$a(n+p) \ = \ v(L_r)\, B^{|w(n)|+|R_r|} + a(n)\, B^{|R_r|} + v(R_r),$$

with $|w(n)|$ growing by $|L_r| + |R_r|$ every $p$ steps.

## 3 The bound

In the shift $T = S^p$ that is a first-order recurrence with a geometric forcing term and a constant, killed by $(T - B^{|R_r|})(T - B^{|L_r|+|R_r|})(T - 1)$. Taking the DISTINCT roots that occur across the residue classes, pulling each back to $S^p - \lambda$, and allowing the pre-period to raise the numerator's degree gives a monic annihilator of order $S = 5$. The earlier figure, $p+3$, was the one-dimensional count carried over unchanged; with $p$ classes carrying $p$ different pairs it is too small, and a residual test that trusts it checks too few coefficients to prove anything.

## 4 The proof

**Lemma 1.** *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^3 - \sum_i c_i z^{3-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of $a$, so $A$ annihilates $r$ as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and $S$ consecutive zeros of $r$ force $r$ to vanish at every later index.*

*Proof.* $A$ annihilates $a$, hence every shift of $a$, hence any linear combination of shifts; that is $r$. A monic recurrence with nonzero constant term determines each term from the $S$ before it and each from the $S$ after it, so a run of $S$ zeros propagates. □

**Theorem 2.**

$$a(n) \;=\; a(n-1) + 16\,a(n-2) - 16\,a(n-3)$$

*for every $n > 3$.*

*Proof.* The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by $A$ where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 3+1$ onwards, and the run exceeds $S+3$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 3$ the recurrence fails on the entry's own published terms, so no larger range holds. □

## 5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 25 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator $A$ was itself tested on terms it had not been given: the model was evaluated beyond the $S$ values that determine it and $A$ reproduced them. A bound that is too small fails this test, which is why it is run.

## References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A263245.

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[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.

[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.